Venkata Lakshmi Parimala Pasupuleti

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Virginia, USA (remote; open to SF relocation)

PROFESSIONAL SUMMARY

Software engineer with 9+ years building data-intensive systems and AI agents in production. Deployed Anthropic Claude into financial data pipelines at Intuit -- grounding agent answers in customer financial records and catching hallucinations with CI eval harnesses before release. Built **Leasel**, an 8-agent LangGraph HITL platform that automates financial compliance investigation: retrieves evidence, generates resolution paths cited to source, escalates to humans when uncertain. 7 years at Entrata owning backend systems that processed 180K+ regulated financial records annually with zero errors. Strong in TypeScript, Python, PostgreSQL, Redis, and AWS. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Case studies: Leasel · AgentProbe

CORE COMPETENCIES

AI Agents (LangGraph, production) | Financial Data Grounding | LLM Evaluation / Evals | TypeScript / Node.js | Python (9+ yrs) | PostgreSQL / Redis | Async Processing at Scale | AWS (EC2 / S3 / SQS / Lambda) | Legacy System Integration | Data-Intensive Backends | End-to-End Feature Ownership | HITL Escalation Design

EXPERIENCE

Intuit

Jun 2026 - Present

Software Engineer · Virginia, USA

- Deploy Anthropic Claude into production financial data pipelines -- ground agent answers in customer financial records so every response traces back to source, catching the subtle errors that arise when models reason over structured financial data.
- Build CI eval harnesses (LLM-as-judge + golden datasets) that block hallucinations and quality regressions before every release -- the safety layer that makes financial agents trustworthy.
- Design structured-output schemas and retrieval pipelines; add guardrails around LLM calls (input validation, sensitive-data handling, output checks before downstream consumption).
- Build FastAPI backends and React/TypeScript front ends shipped via GitHub Actions CI/CD to AWS; ML pipelines with scikit-learn and XGBoost from ingestion through prediction APIs.

Entrata

Oct 2020 - Jul 2024

Senior Software Engineer · Multi-tenant SaaS, 100K+ monthly active users

- Owned data-intensive financial compliance backend: annual federal data import processing 180,000+ records with validation, reconciliation, and rollback -- zero errors across 6 cycles. Same category of authoritative-source grounding Agave's agents require.
- Built RabbitMQ async processing for rent calculations across thousands of tenant records; retry logic and idempotency eliminated double-processing -- distributed queuing at production scale.
- Integrated with legacy federal APIs (HUD TRACS, state housing systems) that had minimal documentation -- reading from systems not designed to be queried, analogous to Agave's on-prem challenge.
- Optimized 200+ SQL queries (30% performance gain); owned AWS S3 PHI migration with Libsodium encryption (100% compliance); led Docker microservices redesign (50% better scalability).
- Delivered 20+ features end-to-end (Python, TypeScript, Java, React, PostgreSQL) across 120+ Agile sprints; owned production incident response.

Entrata India

Sep 2018 - Oct 2020

Junior Software Engineer · Pune, India

- Built data import and reconciliation workflows for 1.6M HUD/Tax Credit records; property management modules in Python, PHP, Java, PostgreSQL; led AWS S3 migration.

PROJECTS

Leasel -- Financial Compliance AI Agent

github.com/venkatalakshmiparimala/Lease_complaince_rag

8-agent LangGraph HITL system for financial compliance automation: classifies issues, retrieves relevant rules and case history, generates evidence-backed resolution paths cited to source, escalates to humans when confidence is low. Hand-rolled hybrid RAG (TF-IDF + semantic reranking). Core design rule: LLM reasons, humans decide every financial/regulatory call. LLM-as-judge CI regression harness keeps agent quality from drifting as data changes.

Python · LangGraph (8-agent) · FastAPI · Streamlit · Hybrid RAG · SQLite · CI/CD · LLM-as-Judge · HUD API

AgentProbe -- Open-Source LLM Evaluation Toolkit

github.com/venkatalakshmiparimala/AgentProbe

15-endpoint FastAPI backend routing across 5 LLM providers. 5-metric LLM-as-judge eval engine: factual accuracy, hallucination rate, retrieval quality, completeness, compliance. CI regression tests on golden datasets -- the infrastructure needed to catch agent quality regressions when agents act on customer financial data. Runs 100% locally at zero cost.

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · Anthropic · OpenAI · Groq · Ollama

AI on FHIR -- Clinical NLP & LLM API Integration

github.com/venkatalakshmiparimala/AI_on_FHIR

LLM + BERT + embeddings translate natural language questions into FHIR API calls against clinical data records -- reduced manual lookup time by ~60%. HIPAA-compliant role-based access; Docker containerized.

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026

GPA 3.87 · NLP pipelines, fine-tuning (HuggingFace + PyTorch), anomaly detection, regression modeling (R^2=0.977)

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017

TECHNICAL SKILLS

AI & Agents: LLM APIs (Anthropic Claude, OpenAI, Groq, Ollama), LangGraph, LangChain, RAG Pipelines, Hybrid Search, Embeddings, Prompt Engineering, Multi-Agent Orchestration, HITL, LLM-as-Judge Evaluation, Fine-tuning (HuggingFace, PyTorch), MCP

Languages & Web: Python (9+ yrs), TypeScript, JavaScript, Node.js, React, Java, Spring Boot, PHP, SQL, Bash

Databases & Data: PostgreSQL, MySQL, Redis, MongoDB, SQLite, Snowflake, ETL Pipelines, Query Optimization, PySpark

Frameworks & APIs: FastAPI, Flask, Django, Express, REST APIs, GraphQL, RabbitMQ, Kafka, Celery

Cloud & DevOps: AWS (S3, EC2, Lambda, Glue, SQS, API Gateway, IAM, CDK), Azure (OpenAI, AKS), GCP (Vertex AI), Docker, Kubernetes, Terraform, GitHub Actions CI/CD

ML: scikit-learn, XGBoost, PyTorch, ChromaDB, Pinecone, BERT, spaCy, HuggingFace, Prometheus, Grafana